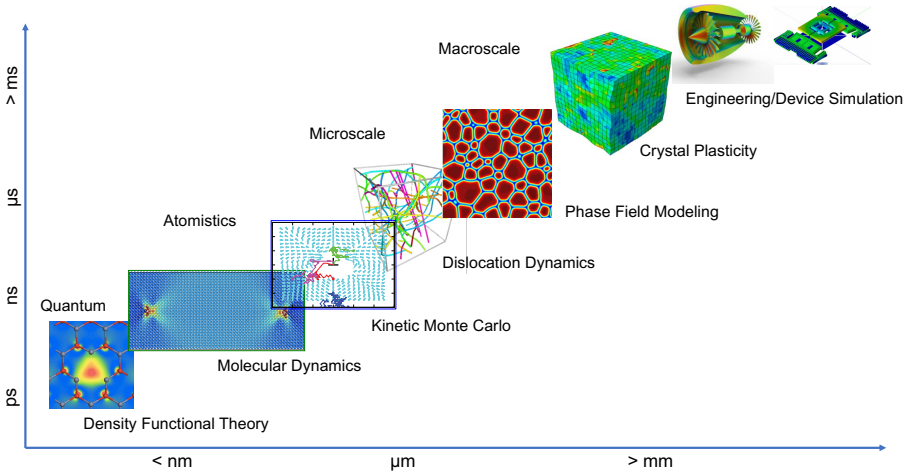
Introduction to Differential Equations

Governing Physical Law to Differential Equation to Materials Simulation

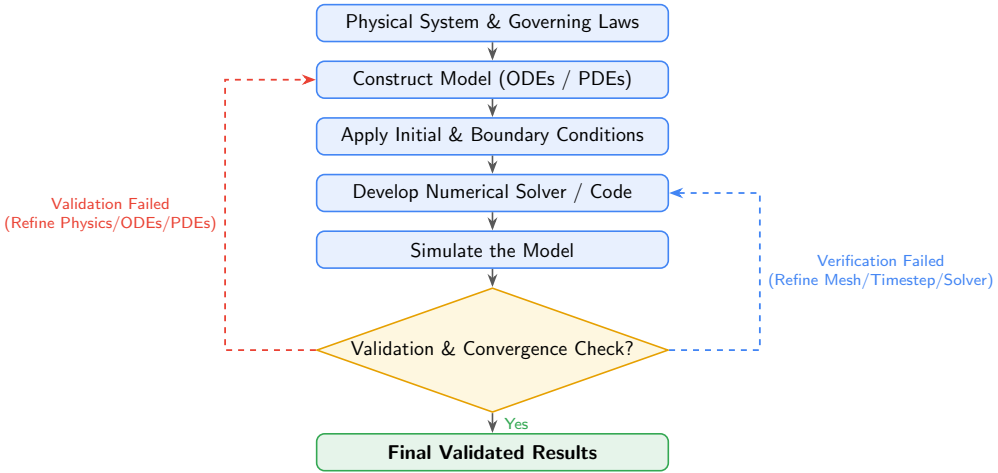
Anuj Goyal

Department of Materials Science and Metallurgical Engineering
Indian Institute of Technology Hyderabad

Materials Modelling across various length and time scales



Stages in Development of a Numerical Model



Ordinary Differential Equation

Differential equation of order 'n' (highest derivative):

$$a_n(t)\frac{d^n y}{dt^n} + a_{n-1}(t)\frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1(t)\frac{dy}{dt} + a_0(t)y = g(t)$$

Where:

- t is the independent variable
- $y(t)$ is the dependent variable (the state/unknown function)
- $a_n(t), a_{n-1}(t), \ldots, a_0(t)$ are coefficient functions that depend only on t (or are constant numbers)
- $g(t)$ is the driving/forcing function that depends only on t

n-th order linear ordinary differential equation (ODE):

$$a_n(t)\frac{d^n y}{dt^n} + a_{n-1}(t)\frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1(t)\frac{dy}{dt} + a_0(t)y = g(t)$$

2-nd order linear partial differential equation (PDE):

$$A(x,y)\frac{\partial^2 u}{\partial x^2} + B(x,y)\frac{\partial^2 u}{\partial x \partial y} + C(x,y)\frac{\partial^2 u}{\partial y^2} + D(x,y)\frac{\partial u}{\partial x} + E(x,y)\frac{\partial u}{\partial y} + F(x,y)u = g(x,y)$$

Linear vs. Non-Linear ODEs

An n-th order Ordinary Differential Equation is called linear if the dependent variable $y(t)$ and all its derivatives $(y', y'', \ldots, y^{(n)})$ appear only to the first power and are never multiplied together or placed inside non-linear functions.

Examples:

- Linear: Forced damped harmonic oscillator: $m\frac{d^2y}{dt^2} + b\frac{dy}{dt} + ky = F_0\cos(\omega t)$
- Non-linear: Simple pendulum: $\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin(\theta) = 0$

Differential Equations: General Implicit Forms

Ordinary differential equation (ODE):

$$F\left(t, y, \frac{dy}{dt}, \frac{d^2y}{dt^2}, \ldots, \frac{d^ny}{dt^n}\right) = 0$$

Partial differential equation (m independent variables):

$$F\left(x_1, x_2, \ldots, x_m,\ u,\ \frac{\partial u}{\partial x_1}, \ldots, \frac{\partial u}{\partial x_m}, \frac{\partial^2 u}{\partial x_1^2}, \frac{\partial^2 u}{\partial x_1 \partial x_2}, \ldots, \frac{\partial^n u}{\partial x_m^n}\right) = 0$$

The Explicit n-th Order Form

Isolated Derivative Definition

$$\frac{d^n y}{dt^n} = f\left(t,\, y,\, \frac{dy}{dt},\, \frac{d^2 y}{dt^2},\, \ldots,\, \frac{d^{n-1} y}{dt^{n-1}}\right)$$

The highest derivative is isolated and given explicitly as a function of the current state.

- Calculates the immediate rate of change directly from known conditions at time t.
- Serves as the direct input format required by standard step-by-step solvers (e.g., Euler, RK4).

An n-th order ODE requires n conditions to find a unique, specific solution

Example: 2nd Order ODE

$$\frac{d^2y}{dt^2} = 6t$$

Integrate once with respect to t:

$$\frac{dy}{dt} = \int 6t\, dt = 3t^2 + C_1$$

Integrate a second time to find $y(t)$:

$$y(t) = \int (3t^2 + C_1)\, dt = t^3 + C_1 t + C_2$$

The General Solution:

$$y(t) = t^3 + C_1 t + C_2$$

Because C_1 and C_2 can be any real numbers, this single formula represents an infinite family of curves

Initial Value Problem (IVP)

- All n conditions are specified at the starting instant in the independent variable ($t = t_0$)
- We predict how system evolves forward into time ($t \geq t_0$).
- Example:

$$y'' + p(t)y' + q(t)y = r(t)$$

- Conditions:

$$y(t_0) = y_0, \quad y'(t_0) = v_0$$

- Physically modelling: Evolution, motion, trajectory

Boundary Value Problem (BVP)

- The n conditions are specified at two or more distinct points of the independent variable (usually spatial boundaries $x = a$ and $x = b$
- We predict for example an equilibrium state everywhere inside that domain
- Example:

$$y'' + p(x)y' + q(x)y = r(x)$$

- Conditions:

$$y(a) = y_a, \quad y(b) = y_b$$

- Physically modelling: Deformation, equilibrium, spatial profiles

Solvability: Linear vs. Non-Linear ODEs

Linear ODEs

- An n-th order linear homogeneous ODE possesses n linearly independent solutions $\{y_1(t), y_2(t), \ldots, y_n(t)\}$.
- These functions form a **basis** for an n-dimensional solution vector space.
- **Superposition Principle** $y(t) = c_1 y_1(t) + c_2 y_2(t) + \cdots + c_n y_n(t)$

Non-Linear ODEs

- **Superposition Fails:** Linear combinations of solutions are generally *not* solutions
- **No Basis Representation:** No finite set of basis functions exists to construct all solutions.

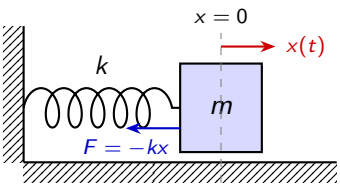
Some of the Differential Equations in Different Fields of Physics

Studying a phenomenon in one field in detail can allow us to extend our knowledge to another field.

System	Governing Law	Equation	Type
Harmonic oscillator	Newton's + Hooke's	$m\frac{d^2x}{dt^2} + kx = 0$	ODE
Quantum	de Broglie's + energy	$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V\psi = E\psi$	ODE
Reaction kinetics	Rate law	$\frac{d[A]}{dt} = -k[A]$	ODE
Electrostatics	Gauss's law	$\nabla^2 V = -\frac{\rho}{\varepsilon_0}$	PDE
Heat conduction	Fourier's + energy cons.	$\frac{\partial T}{\partial t} = \alpha\frac{\partial^2 T}{\partial x^2}$	PDE
Diffusion	Fick's + mass cons.	$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$	PDE

Harmonic Oscillator to MD Simulations

A Mass on a Spring: Ordinary Differential Equations



Displace the mass by $x(t)$ from equilibrium. Two facts close the loop:

1. Hooke's law — the spring pulls back:

$$F = -k\,x.$$

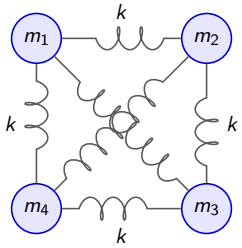
2. Newton's second law — force sets acceleration:

$$F = m\,\ddot{x}.$$

Equation of motion

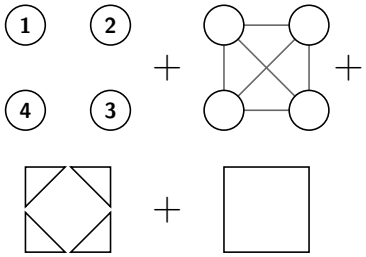
$$m\ddot{x} = -kx \quad \Longleftrightarrow \quad m\frac{d^2x}{dt^2} + kx = 0.$$

Coupled ODEs in MD Simulation



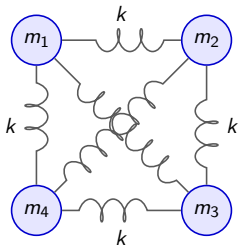
$$m_i \frac{d^2\mathbf{r}_i}{dt^2} = -\nabla_{\mathbf{r}_i} U(\mathbf{r}_1, \ldots, \mathbf{r}_N)$$

Hierarchy of Atomic Interactions



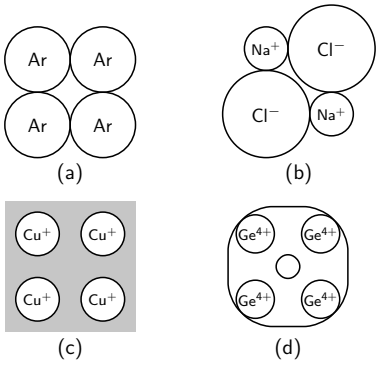
$$U = \sum_{i=1}^{N} v_1(\mathbf{r}_i) + \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N}{}' \phi_{ij}(\mathbf{r}_i, \mathbf{r}_j) + \frac{1}{6} \sum_{i=1}^{N} \sum_{j=1}^{N} \sum_{k=1}^{N}{}' v_3(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) + \cdots$$

Coupled ODEs in MD Simulation



$$m_i \frac{d^2\mathbf{r}_i}{dt^2} = -\nabla_{\mathbf{r}_i} U(\mathbf{r}_1, \ldots, \mathbf{r}_N)$$

Bonding in Atomic Solids



Initial and Boundary Conditions

Initial Conditions

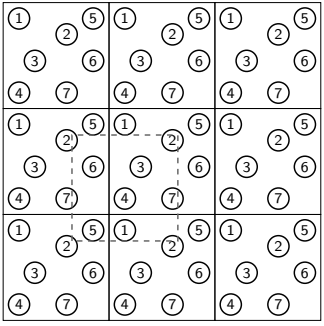
Positions, velocities, and net momentum

$$\mathbf{r}_i(0) = \mathbf{r}_{i,\text{lattice}}$$

$$P(v_{i\alpha}) = \sqrt{\frac{m_i}{2\pi k_B T}} \exp\left(-\frac{m_i v_{i\alpha}^2}{2k_B T}\right)$$

$$\sum_{i=1}^{N} m_i \mathbf{v}_i(0) = \mathbf{0}$$

Periodic Boundary Conditions



Interactions, Differential Equations, and Solvers in MD simulation

1. Interatomic Potential Energy

$$U = \sum_{i=1}^{N} v_1(\mathbf{r}_i) + \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}{}' \phi_{ij}(\mathbf{r}_i, \mathbf{r}_j) + \frac{1}{6}\sum_{i=1}^{N}\sum_{j=1}^{N}\sum_{k=1}^{N}{}' v_3(\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k) + \cdots$$

2. Force Calculation

$$\mathbf{F}_i = -\nabla_i U(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) = -\nabla_i U(\mathbf{r}^N)$$

3. Equations of Motion (The ODEs)

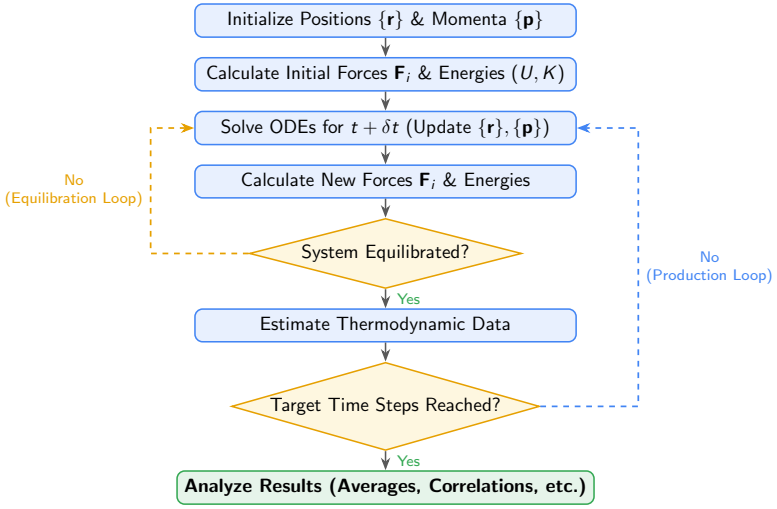
$$\frac{d^2\mathbf{r}_i}{dt^2} = \frac{1}{m_i}\mathbf{F}_i = \frac{1}{m_i}\sum_{j\neq i}\mathbf{f}_{ij}(r_{ij})$$

4. Numerical Solvers (Time Integration)

$$\mathbf{v}_i(t+\delta t) = \mathbf{v}_i(t) + \mathbf{a}_i(t)\delta t$$

$$\mathbf{r}_i(t+\delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t)\delta t + \frac{1}{2}\mathbf{a}_i(t)\delta t^2$$

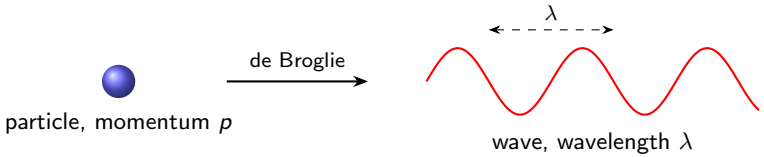
Typical MD Simulation Flowchart



Schrödinger Equation to DFT calculations

Matter Waves: the de Broglie Hypothesis

de Broglie proposed, then particles such as electrons behave as waves — with a wavelength set by their momentum.



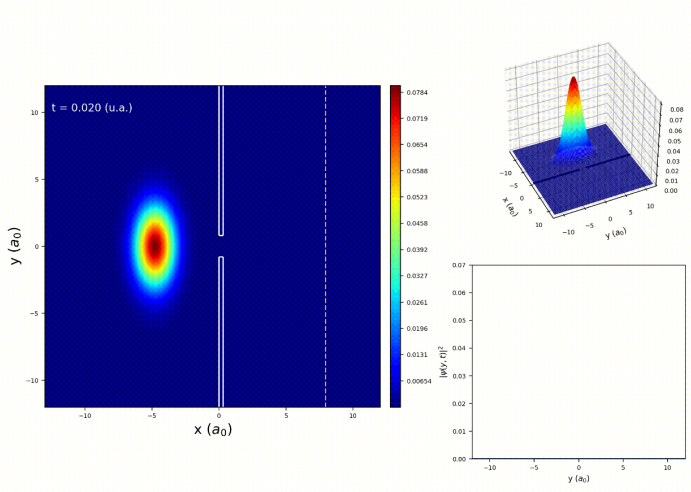
The de Broglie Relationship

$$\lambda = \frac{h}{p}$$

Momentum and Wave Number

$$p = \frac{h}{\lambda} = \frac{h}{2\pi} \cdot \frac{2\pi}{\lambda} \qquad \Longrightarrow \qquad \boxed{p = \hbar k}.$$

Electron Diffraction: Gaussian Wave Packet



F. Desrochers, *Electron Diffraction*. GitHub repository.
Animation source: `https://github.com/FelixDesrochers/Electron-diffraction`

The Time-Independent Schrödinger Equation

Free particle as a wave of spatial frequency k:

$$\psi(x) = A\, e^{ikx}$$

$$\frac{d\psi}{dx} = ik\,\psi$$

$$\frac{d^2\psi}{dx^2} = -k^2\psi$$

The total energy is kinetic plus potential:

$$E = \frac{p^2}{2m} + U(x)$$

$$E = \frac{\hbar^2 k^2}{2m} + U(x)$$

$$k^2 = \frac{2m}{\hbar^2}\big(E - U(x)\big)$$

$$\underbrace{\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x)\right)}_{\text{Hamiltonian } \mathcal{H}}\psi(x) = E\,\psi(x)$$

Schrödinger Equation: ODEs and PDEs

Time-Independent Schrödinger Equation (Many-body)

$$\left[-\sum_i \frac{\hbar^2}{2m_e}\nabla_i^2 - \sum_I \frac{\hbar^2}{2M_I}\nabla_I^2 + \frac{1}{2}\sum_{i\neq j}\frac{e^2}{4\pi\epsilon_0}\frac{1}{|\mathbf{r}_i-\mathbf{r}_j|} + \frac{1}{2}\sum_{I\neq J}\frac{e^2}{4\pi\epsilon_0}\frac{Z_I Z_J}{|\mathbf{R}_I-\mathbf{R}_J|} - \sum_{i,I}\frac{e^2}{4\pi\epsilon_0}\frac{Z_I}{|\mathbf{r}_i-\mathbf{R}_I|}\right]\Psi = E_{\text{tot}}\Psi$$

- 1D system is a **2nd-order ODE**; 3D and many-body systems are **PDEs**.
- Bound states yield **standing waves** with **quantized** discrete energies

Time-Dependent Schrödinger Equation

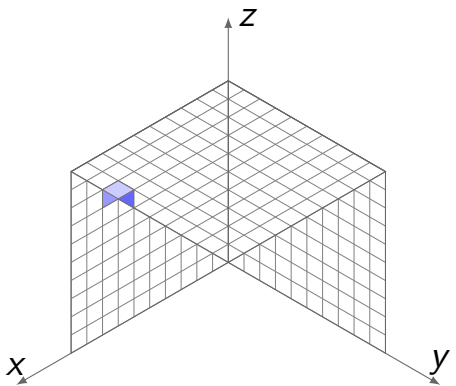
$$i\hbar\frac{\partial\Psi(\mathbf{r},t)}{\partial t} = \hat{H}\Psi(\mathbf{r},t)$$

- **Partial Differential Equation (PDE)** in space and time.
- Yields **traveling waves** / wave packets.

Exponential Wall

Storage requirements for the many-body wavefunction of one unit cell of silicon in the diamond structure

- $\Delta x \sim 0.1$ Å
- $a = 5.43$ Å
- $N_p = (a^3/4)/(\Delta x)^3 \sim 40,000$
- 8 valence electrons per unit cell
- $\Psi = 40,000^8$ complex numbers



Schrodinger Equation to Kohn-Sham (DFT) Equations

Convention: $e = \hbar = m_e = 1$

$$\left[-\frac{1}{2}\nabla^2 + V_{\mathrm{n}}(\mathbf{r}) + V_{\mathrm{H}}(\mathbf{r}) + V_{xc}(\mathbf{r})\right]\phi_i(\mathbf{r}) = \varepsilon_i\phi_i(\mathbf{r})$$

- A PDE solving for time-independent ground-state properties.
- Maps the many-body Hamiltonian to a functional of the electron charge density $\rho(\mathbf{r})$, which is a 3D scalar field.
- Replaces interacting electrons with fictitious single-particle wavefunctions (ϕ_i) that yield the exact same ground-state charge density.
- Reduces the computational storage from $40,000^8$ complex numbers down to $8 \times 40,000$ single-particle grid values.

DFT Calculation Flowchart

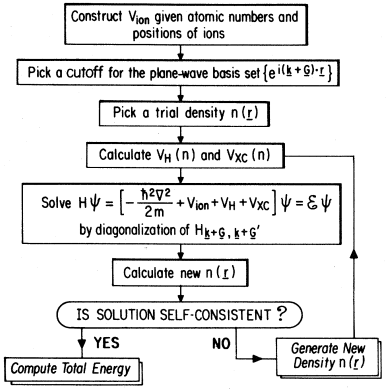


FIG. 7. Flow chart describing the computational procedure for the calculation of the total energy of a solid, using conventional matrix diagonalization.

M. C. Payne, M. P. Teter, D. C. Allan, T. A. Arias, and J. D. Joannopoulos, *Iterative minimization techniques for ab initio total-energy calculations: molecular dynamics and conjugate gradients*, Reviews of Modern Physics **64**, 1045 (1992).

Microscopic to Macroscopic Models

Gauss's Law to Poisson Equation to Charge Density

1. **Gauss's Law (Integral Form)**

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = \frac{1}{\varepsilon_0} \int_V \rho \, d\tau$$

2. **Applying Divergence Theorem**

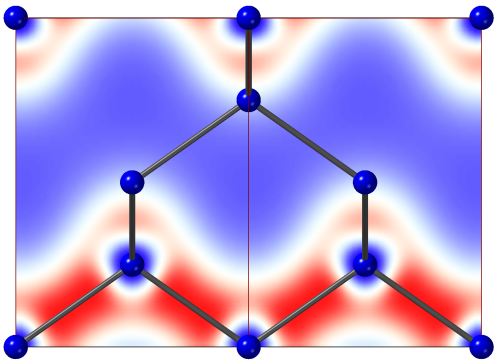
$$\int_V (\nabla \cdot \mathbf{E}) \, d\tau = \frac{1}{\varepsilon_0} \int_V \rho \, d\tau$$

3. **Differential Form (PDE)**

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

4. **Potential from Charge Density ($\mathbf{E} = -\nabla V$)**

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0} \quad \text{(Poisson's Equation)}$$



DFT codes use Poisson Solvers to estimate Hartree Potential $V_H(\mathbf{r})$ from the electron charge density $\rho(\mathbf{r})$

Poisson's Equation: From 3D PDE to 1D ODE

Full 3D Partial Differential Equation

$$\nabla^2 V(x,y,z) = -\frac{\rho(x,y,z)}{\varepsilon_0}$$

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho(x,y,z)}{\varepsilon_0}$$

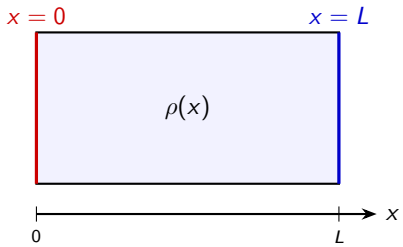
- V depends on three spatial dimensions (x, y, z).
- Requires boundary conditions across all three spatial dimensions (e.g., all 6 faces of a 3D unit cell in x, y, z)
- 3D electron charge density in a crystal unit cell.

1D Ordinary Differential Equation

$$\frac{d^2 V}{dx^2} = -\frac{\rho(x)}{\varepsilon_0}$$

- Partial derivatives reduce to total derivatives $\frac{d^2}{dx^2}$.
- Requires only two boundary conditions at endpoints $x = 0$ and $x = L$.

Boundary Conditions for 1D Electrostatic BVP

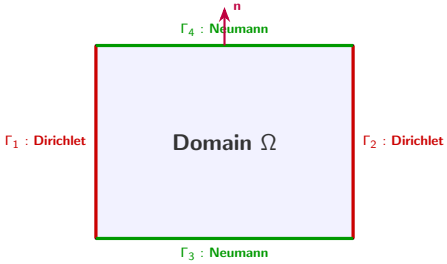


Governing 1D BVP:

$$\frac{d^2V}{dx^2} = -\frac{\rho(x)}{\varepsilon_0}, \quad x \in [0, L]$$

1. **Dirichlet (Fixed Potentials)**
   - $V(0) = V_0, \quad V(L) = V_L$
2. **Neumann (Fixed Electric Field)**
   - $\frac{dV}{dx}\big|_{x=0} = -E_0$
3. **Periodic (Bulk Crystal)**
   - Wrapping boundaries to create an infinitely repeating lattice
   - $V(0) = V(L), \quad \frac{dV}{dx}\big|_{x=0} = \frac{dV}{dx}\big|_{x=L}$

Applying Boundary Conditions to 2D PDE



Governing 2D PDE (Poisson):

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = -\frac{\rho(x,y)}{\varepsilon_0} \quad \text{in } \Omega$$

To solve a 2D PDE, boundary conditions must be specified on all closed boundary edges $\partial\Omega = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_4$.

1. Dirichlet Boundary Condition

- Specifies the **value** of potential V
- Fixed potential / temperature:

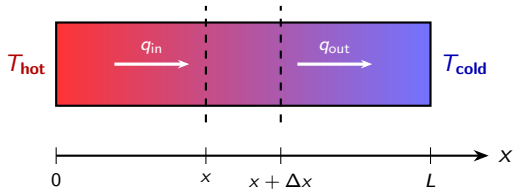
$$V(x,y) = V_{\text{fixed}}(x,y) \quad \text{on } \Gamma_D$$

2. Neumann Boundary Condition

- Specifies the **normal derivative** (flux)
- Flux / gradient perpendicular to boundary:

$$\frac{\partial V}{\partial n} = \mathbf{n} \cdot \nabla V = g(x,y) \quad \text{on } \Gamma_N$$

Fourier's Law to Heat Equation



Mesoscale and macroscale models treat materials as continuous media rather than discrete atoms, using fields like temperature and gradients to solve the macroscopic heat equation.

1. Fourier's Law

$$q = -k \frac{\partial T}{\partial x}$$

2. Energy Balance

$$\big(q(x) - q(x+\Delta x)\big)A = \frac{\partial Q}{\partial t} = \rho c_p A \, \Delta x \, \frac{\partial T}{\partial t}$$

$$-\frac{\partial q}{\partial x} = \rho c_p \frac{\partial T}{\partial t}$$

3. Heat Equation (PDE)

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right)$$

How an ODE BVP Emerges from a PDE at Steady State

Heat flow in a 1D metal rod $T(x,t)$:

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right)$$

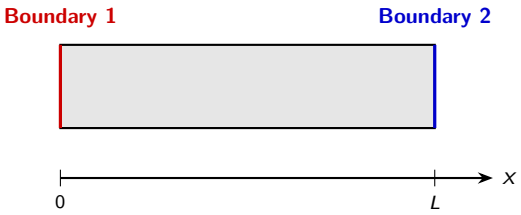
$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

- At equilibrium the time derivative drops to zero:

$$\frac{\partial T}{\partial t} = 0 \implies \frac{d^2 T}{dx^2} = 0$$

- x is now the only independent variable left.
- left with an ODE Boundary Value Problem (BVP) to find the steady state temperature $T(x)$ along the rod

Applying Boundary Conditions to the ODE



Governing Equation:

$$\frac{d^2T}{dx^2} = 0 \quad \text{for } 0 < x < L$$

To solve this 2nd-order ODE, we must specify exactly **two** independent conditions at the domain boundaries.

1. Dirichlet Boundary Condition

- Specifies the **value** of the variable
- Temperatures at the ends:

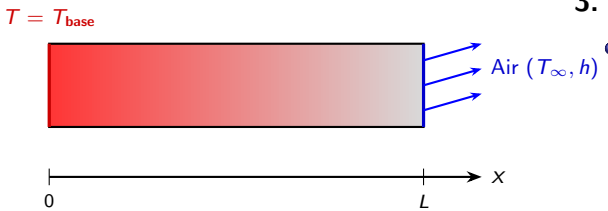
$$T(0) = T_{\text{hot}}, \quad T(L) = T_{\text{cold}}$$

2. Neumann Boundary Condition

- Specifies the **derivative** (gradient)
- Heat flux at the ends:

$$-k \left.\frac{dT}{dx}\right|_{x=0} = q_{\text{in}}$$

Mixed and Robin Boundary Conditions



A 2nd-order ODE needs **two** conditions, but they do not have to be of the same type! Real materials often have asymmetric boundary physics.

3. Robin Boundary Condition

- A linear combination of T and flux at the *same* boundary:

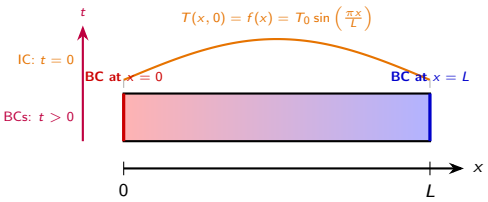
$$-k \left.\frac{dT}{dx}\right|_{x=L} = h\big(T(L) - T_\infty\big)$$

4. Mixed Boundary Conditions

- Fixed T (*Dirichlet*) at one end and flux (*Neumann*) at the other:

$$T(0) = T_{\text{base}}, \quad \left.\frac{dT}{dx}\right|_{x=L} = 0$$

Initial and Boundary Conditions



Governing 1D Heat Equation:

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad \text{for } 0 < x < L,\ t > 0$$

- **1st-order in time (t):** Requires **1** Initial Condition across domain.
- **2nd-order in space (x):** Requires **2** Boundary Conditions at domain ends.

Initial Condition (IC) at $t = 0$

- Specifies spatial temperature profile at start:

$$T(x,0) = f(x) \quad \text{for } 0 \le x \le L$$

Boundary Conditions (BCs) for $t > 0$

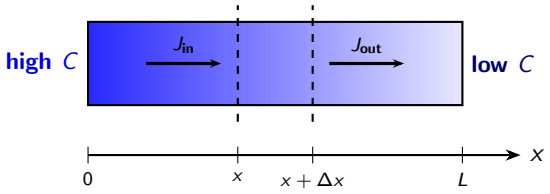
Dirichlet (Fixed Temperature)

$$T(0,t) = T_A, \quad T(L,t) = T_B$$

Neumann (Specified Flux / Insulated)

$$-k \left.\frac{\partial T}{\partial x}\right|_{x=0} = q_{\text{in}}, \quad \left.\frac{\partial T}{\partial x}\right|_{x=L} = 0 \quad \text{(Insulated)}$$

Fick's Law to Diffusion Equation



Macroscale and mesoscale models describe mass transport using continuous composition fields. They solve governing conservation laws (like the Cahn-Hilliard or Allen-Cahn equations) driven by chemical potential gradients, deriving thermodynamic forces from a free-energy functional without tracking individual atomic trajectories.

1. Fick's First Law

$$J = -D\,\frac{\partial C}{\partial x}$$

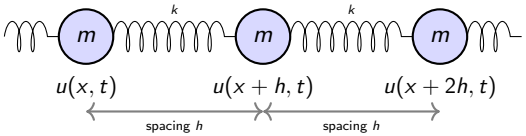
2. Mass Balance

$$\big(J(x) - J(x+\Delta x)\big)A = \frac{\partial N}{\partial t} = A\,\Delta x\,\frac{\partial C}{\partial t}$$

$$-\frac{\partial J}{\partial x} = \frac{\partial C}{\partial t}$$

3. General Diffusion Equation (PDE)

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x}\left(D\frac{\partial C}{\partial x}\right)$$



Hooke's Law

Net restoring force acting on the mass at location $(x + h)$:

$$F_{\text{Hooke}} = k[u(x + 2h, t) - u(x + h, t)] - k[u(x + h, t) - u(x, t)]$$

Deriving the Wave Equation

Applying Newton's 2nd Law ($ma = F_{\text{Hooke}}$)

$$m\frac{\partial^2 u(x+h,t)}{\partial t^2} = k\left[u(x+2h,t) - 2u(x+h,t) + u(x,t)\right]$$

Rearranging for Acceleration

$$\frac{\partial^2 u(x+h,t)}{\partial t^2} = \frac{k}{m}\left[u(x+2h,t) - 2u(x+h,t) + u(x,t)\right]$$

Scaling to Continuum Parameters

- For a chain of N masses over total length $L = Nh$ and total mass $M = Nm$ with total spring constant $K = k/N$:

$$\frac{\partial^2 u(x+h,t)}{\partial t^2} = \left(\frac{KL^2}{M}\right)\frac{u(x+2h,t) - 2u(x+h,t) + u(x,t)}{h^2}$$

Deriving the Wave Equation

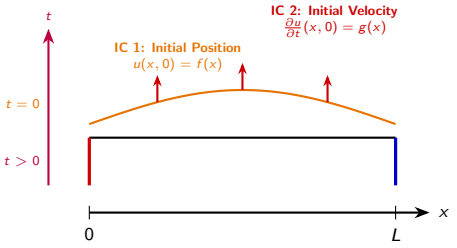
Using the standard definition of the second spatial derivative:

$$\lim_{h \to 0} \frac{u(x+2h,t) - 2u(x+h,t) + u(x,t)}{h^2} = \frac{\partial^2 u(x+h,t)}{\partial x^2}$$

$$\frac{\partial^2 u(x,t)}{\partial t^2} = c^2 \frac{\partial^2 u(x,t)}{\partial x^2}$$

- $u(x,t)$ represents spatial displacement depending on position x and time t.
- $c = \sqrt{\frac{KL^2}{M}}$ (or $\sqrt{T/\rho}$, $\sqrt{E/\rho}$) defines the physical propagation speed of the disturbance.
- Requires **two initial conditions** across the spatial domain at $t = 0$:
  - Initial position profile $u(x,0)$
  - Initial velocity profile $\frac{\partial u}{\partial t}(x,0)$
- Requires **two boundary conditions** at domain ends ($x = 0, L$) for all times $t > 0$.

Initial and Boundary Conditions for Wave PDE



$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{for } 0 < x < L,\ t > 0$$

- **2nd-order in time (t):** Requires **2** Initial Conditions across domain.
- **2nd-order in space (x):** Requires **2** Boundary Conditions at domain ends.

1. Two Initial Conditions (ICs) at $t = 0$

- *Initial Position:*

$$u(x,0) = f(x) \quad \text{for } 0 \le x \le L$$

- *Initial Velocity:*

$$\left.\frac{\partial u}{\partial t}\right|_{t=0} = g(x) \quad \text{for } 0 \le x \le L$$

2. Boundary Conditions (BCs) for $t > 0$

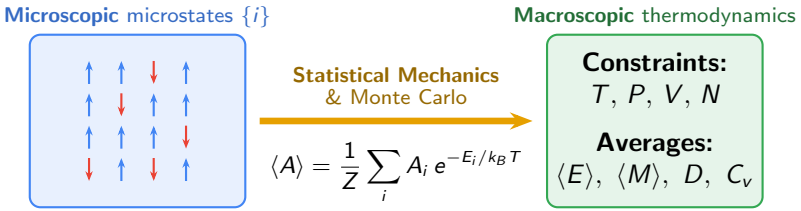
- *Fixed Ends (Dirichlet)*

$$u(0,t) = 0, \quad u(L,t) = 0$$

- *Free / Force-Free Ends (Neumann)*

$$\left.\frac{\partial u}{\partial x}\right|_{x=0} = 0, \quad \left.\frac{\partial u}{\partial x}\right|_{x=L} = 0$$

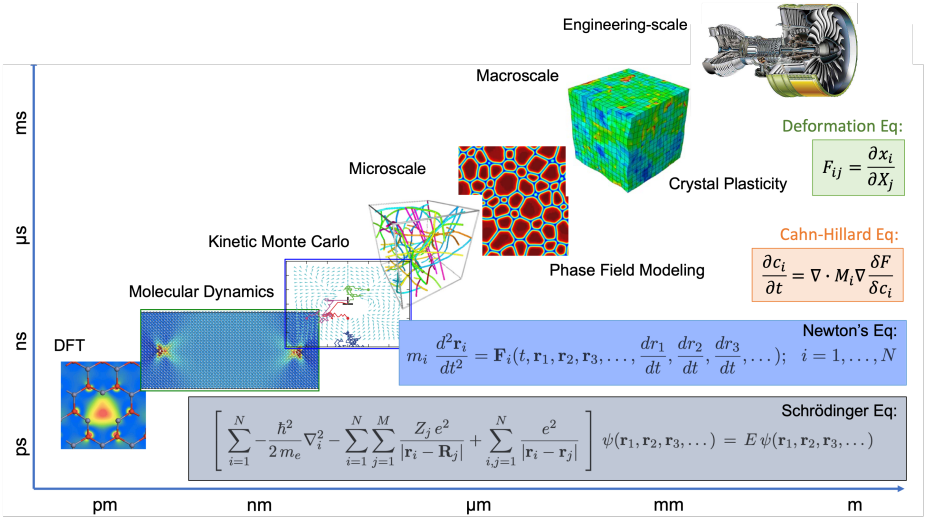
Bridging Microscopic and Macroscopic Domains



- **Equilibrium MC (Metropolis):** Samples equilibrium macrostates from the Boltzmann distribution using the probability $p = \min\left(1,\ e^{-\Delta E/k_B T}\right)$. Used to solve equilibrium problems like Ising magnetism and alloy ordering.
- **Kinetic MC (kMC):** Models non-equilibrium dynamic evolution by integrating the master/rate equations (a system of ODEs). Employ transition rates $r = \nu \, e^{-E_a/k_B T}$ to solve kinetics problems like self-diffusion and defect migration.

Statistical mechanics + Monte Carlo connect atomistic microstate with macroscopic thermodynamic estimate.

Materials Modelling across various length and time scales



Numerical Methods to Solve ODEs and PDEs

A simple 1st order ODE IVP

We study the first-order initial value problem

$$y'(t) = f\big(t, y(t)\big), \qquad y(t_0) = y_0, \qquad t \geq t_0.$$

Example:

$$y' = \lambda y, \quad y(0) = 1 \qquad \Longrightarrow \qquad y(t) = e^{\lambda t}.$$

- How to solve it numerically?
- Approach: march forward and estimate y(t)

Taylor Series methods

First step:

$$y(t_1) = y(t_0 + h) = y_0 + h\,y'(t_0) + \tfrac{h^2}{2!}y''(t_0) + \tfrac{h^3}{3!}y'''(t_0) + \cdots$$

General Taylor expansion (truncated at first order):

$$y_{n+1} = y(t_n + h) = y_n + h\,y'(t_n) + \tfrac{h^2}{2}y''(\xi), \qquad y'(t_n) = f(t_n, y_n).$$

- From the ODE definition, we know:

$$y'(t) = f(t, y(t))$$

- Approximate:

$$y_{n+1} \approx y_n + h\,f(t_n, y_n)$$

- Numerical approximation:

$$u_{n+1} = u_n + h\,f(t_n, u_n)$$

Forward Euler

Idea: truncate the Taylor expansion at first order,

$$y(t_n + h) = y_n + h\, y'(t_n) + \tfrac{h^2}{2} y''(\xi), \qquad y'(t_n) = f(t_n, y_n).$$

Forward Euler (Explicit)

$$u_{n+1} = u_n + h\, f(t_n, u_n)$$

- **Explicit:** the right-hand side uses only the known value y_n
- Geometrically, we step along the tangent slope $f(t_n, u_n)$

Example

Consider an ODE:

$$y' = y + t, \quad y(0) = 1, \quad 0 \leq t \leq 1$$

Exact solution:

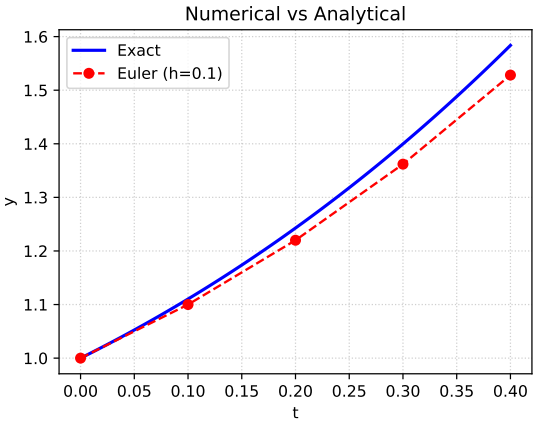
$$y(t) = 2e^t - t - 1$$

Forward Euler in Python

[style=py] import numpy as np
Forward Euler solver function def forward$_euler(f, t0, y0, h, steps) : t = np.zeros(steps + 1)y =$
$np.zeros(steps + 1)t[0] = t0y[0] = y0$
for i in range(steps): y[i+1] = y[i] + h * f(t[i], y[i]) t[i+1] = t[i] + h
return t, y

Forward Euler vs. Exact Solution ($h = 0.1$)

Step n	t_n	Euler y_n	Exact $y(t_n)$	Error
0	0.0	1.0000	1.0000	0.0000
1	0.1	1.1000	1.1103	0.0103
2	0.2	1.2200	1.2428	0.0228
3	0.3	1.3620	1.3997	0.0377
4	0.4	1.5282	1.5836	0.0554



Error Analysis

Exact solution:

$$y(t_{n+1}) = y(t_n) + h\,y'(t_n) + \frac{h^2}{2}y''(\xi_n), \qquad y'(t_n) = f(t_n, y(t_n)).$$

Numerical approximation:

$$u_{n+1} = u_n + hf(t_n, u_n).$$

Error:

$$e_{n+1} = y_{n+1} - u_{n+1} = y_n - u_n + h[f(t_n, y_n) - f(t_n, u_n)] + \frac{h^2}{2}y''(\xi_n)$$

$$e_{n+1} = e_n + \frac{h[f(t_n, y_n) - f(t_n, u_n)]}{y_n - u_n}e_n + \frac{h^2}{2}y''(\xi_n)$$

Using Lipschitz constant $\mathcal{L}$ and bounding $|y''(\xi_n)| \leq K$:

$$|e_{n+1}| \leq |e_n| + h\mathcal{L}|e_n| + \frac{h^2}{2}K = (1 + h\mathcal{L})|e_n| + \underbrace{\frac{h^2}{2}K}_{\text{Local Error } \mathcal{O}(h^2)}$$

Forward Euler: Local vs. Global Error

Global error (truncation error accumulated over N steps):

By solving the recurrence relation $|e_{n+1}| \leq (1 + h\mathcal{L})|e_n| + \frac{h^2}{2}K$, we get:

$$|e_n| \leq \frac{hK}{2\mathcal{L}}\left(e^{\mathcal{L}(t_n - t_0)} - 1\right) = \mathcal{O}(h)$$

Takeaway

The global error is **one order lower** than the local error because we accumulate $\mathcal{O}(h^2)$ errors over $N \propto 1/h$ steps.

Global error (including round-off error ρ):

$$|e_n| \leq \left(\frac{e^{\mathcal{L}(t_n - t_0)} - 1}{\mathcal{L}}\right)\left[\frac{hK}{2} + \frac{\rho}{h}\right].$$

Can we improve on the Forward Euler method?

The answer to this opens up various numerical methods, primarily designed to solve initial-value problems.

Numerical Methods to Solve BVPs

Main Numerical Methods for BVPs

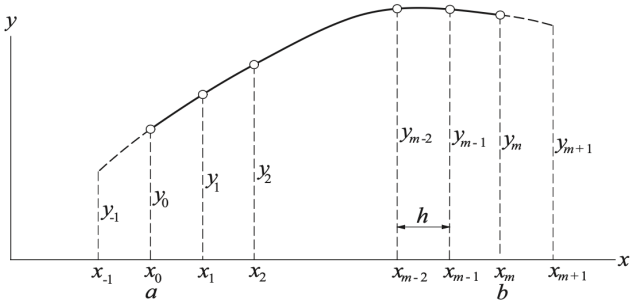
- Finite Difference Method (FDM)
- Variational / Finite Element Method (FEM)
- Spectral Methods

All continuous discretization techniques (FDM, FEM, Spectral) convert the differential operator into matrix operations (Linear Algebra):

- Linear Systems ($A\mathbf{x} = \mathbf{b}$)
- Eigenvalue Problems ($A\mathbf{x} = \lambda\mathbf{x}$)
- Non-Linear Systems ($\mathbf{F}(\mathbf{x}) = \mathbf{0}$)

Finite Difference Method: Discretization

- The fundamental approach is to divide the range of integration (a, b) into m equal subintervals of length h.
- The values of the numerical solution at the discrete mesh points are denoted by y_i, where $i = 0, 1, \ldots, m$.



Finite Difference Method: Approximating Derivatives

- We approximate the true continuous derivatives of y by replacing them with finite difference expressions (typically central differences):

$$y_i' = \frac{y_{i+1} - y_{i-1}}{2h}$$

$$y_i'' = \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2}$$

- The original differential equation is now enforced **only at the discrete mesh points**.
- As a result, the differential equation is completely replaced by $m + 1$ simultaneous algebraic equations, where our unknowns are y_i.

Finite Difference Method: Systems & Accuracy

- **Linear ODE:** The resulting equations are linear. We assemble and solve a matrix system:

$$Ay = b$$

- **Nonlinear ODE:** The resulting equations are nonlinear. We form a residual system and solve it iteratively (e.g., via Newton-Raphson):

$$F(y) = 0$$

Accuracy Considerations:

- The truncation error in standard first central difference approximations is $O(h^2)$.
- Consequently, FDM is generally less accurate for a given step size than the Shooting Method ($O(h^4)$ RK4 integrators).
- When solving nonlinear FDM systems, do not set an overly strict Newton-Raphson convergence tolerance; it should realistically reflect the larger $O(h^2)$ discretization error.

Finite Difference Method: General Setup

The General Second-Order BVP:

$$y'' = f(x, y, y')$$

Boundary Conditions (at $x = a$ and $x = b$):

- **Start Boundary ($x = a$):**

$$y(a) = \alpha \quad \text{(Dirichlet)} \quad \text{or} \quad y'(a) = \alpha \quad \text{(Neumann)}$$

- **End Boundary ($x = b$):**

$$y(b) = \beta \quad \text{(Dirichlet)} \quad \text{or} \quad y'(b) = \beta \quad \text{(Neumann)}$$

Finite Difference Method: Discretization

Applying Central Differences:

- Approximating the derivatives at the mesh points $i = 0, 1, \ldots, m$ yields:

$$\frac{y_{i-1} - 2y_i + y_{i+1}}{h^2} = f\left(x_i, y_i, \frac{y_{i+1} - y_{i-1}}{2h}\right)$$

Discretizing the Boundary Conditions:

- At $i = 0$: $y_0 = \alpha$ or $\frac{y_1 - y_{-1}}{2h} = \alpha$
- At $i = m$: $y_m = \beta$ or $\frac{y_{m+1} - y_{m-1}}{2h} = \beta$

- *Note:* The terms y_{-1} and y_{m+1} are associated with "ghost" or "spillover" points located outside the physical domain (a, b). This spillover can be eliminated using the boundary conditions.

Finite Difference Method: The Algebraic System

Rearranging into a System of Equations:

- **At start boundary ($i = 0$):**

$$y_{-1} - 2y_0 + y_1 - h^2 f\left(x_0, y_0, \frac{y_1 - y_{-1}}{2h}\right) = 0$$

- **At interior nodes ($i = 1, 2, \ldots, m-1$):**

$$y_{i-1} - 2y_i + y_{i+1} - h^2 f\left(x_i, y_i, \frac{y_{i+1} - y_{i-1}}{2h}\right) = 0$$

- **At end boundary ($i = m$):**

$$y_{m-1} - 2y_m + y_{m+1} - h^2 f\left(x_m, y_m, \frac{y_{m+1} - y_{m-1}}{2h}\right) = 0$$

Next Step: Apply the given boundary conditions to substitute and eliminate the spillover terms y_{-1} and y_{m+1}.

Finite Difference Method: Final System

We finish with $m+1$ equations for the unknowns $y_0, y_1, \ldots, y_m$:

1. At start Boundary ($i = 0$):

$$\begin{cases} y_0 - \alpha = 0 & \text{if } y(a) = \alpha \\ -2y_0 + 2y_1 - h^2 f(x_0, y_0, \alpha) - 2h\alpha = 0 & \text{if } y'(a) = \alpha \end{cases}$$

2. At interior Nodes ($i = 1, 2, \ldots, m-1$):

$$y_{i-1} - 2y_i + y_{i+1} - h^2 f\left(x_i, y_i, \frac{y_{i+1} - y_{i-1}}{2h}\right) = 0$$

3. At end Boundary ($i = m$):

$$\begin{cases} y_m - \beta = 0 & \text{if } y(b) = \beta \\ 2y_{m-1} - 2y_m - h^2 f(x_m, y_m, \beta) + 2h\beta = 0 & \text{if } y'(b) = \beta \end{cases}$$

Example: 1D Poisson Equation

Model Problem:

$$-u''(x) = f(x) \quad \text{on } [0,1], \quad u(0) = 0,\ u(1) = 0$$

Domain Discretization:

- Divide $[0,1]$ into N equal subintervals of length $h = \frac{1}{N}$.
- Grid points: $x_i = ih$ for $i = 0, 1, \ldots, N$.
- Unknowns: Internal grid values $U_1, U_2, \ldots, U_{N-1}$.

Central Difference Approximation:

- Replace $-u''(x_i)$ at internal nodes $i = 1, 2, \ldots, N-1$:

$$\frac{-U_{i-1} + 2U_i - U_{i+1}}{h^2} = f(x_i)$$

- Apply Dirichlet boundary conditions: $U_0 = 0$ and $U_N = 0$.

1D Poisson Equation: Discretization

Try yourself!

1D Poisson Equation: Matrix Assembly

Complete System Matrix (AU = F) of size $(N+1) \times (N+1)$:

$$\frac{1}{h^2}\begin{pmatrix} h^2 & 0 & 0 & \dots & 0 & 0 \\ -1 & 2 & -1 & \dots & 0 & 0 \\ 0 & -1 & 2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 2 & -1 \\ 0 & 0 & 0 & \dots & 0 & h^2 \end{pmatrix} \begin{pmatrix} U_0 \\ U_1 \\ U_2 \\ \vdots \\ U_{N-1} \\ U_N \end{pmatrix} = \begin{pmatrix} 0 \\ f_1 \\ f_2 \\ \vdots \\ f_{N-1} \\ 0 \end{pmatrix}$$

Example: Linear two-point BVP with mixed boundary conditions

The Boundary Value Problem:

$$y'' = -4y + 4x$$

Boundary Conditions:

$$y(0) = 0 \quad \text{and} \quad y'(\pi/2) = 0$$

FDM Parameters:

- Let the number of subintervals be $m = 10$.
- Domain: $[0, \pi/2]$
- Step size: $h = \frac{\pi/2 - 0}{10} = \frac{\pi}{20}$

Identifying the RHS Function:

- From the general form $y'' = f(x, y, y')$, we have:

$$f(x, y, y') = -4y + 4x$$

Nodal Equations

1. Start Node ($i = 0$): Dirichlet boundary condition:

$$y_0 = 0$$

2. Interior Nodes ($i = 1, 2, \ldots, 9$): Substitute $f(x_i, y_i)$ into the general interior equation:

$$y_{i-1} - 2y_i + y_{i+1} - h^2(-4y_i + 4x_i) = 0$$

$$y_{i-1} + (4h^2 - 2)y_i + y_{i+1} = 4h^2 x_i$$

3. End Node ($i = 10$): Neumann boundary condition:

$$2y_9 - 2y_{10} - h^2(-4y_{10} + 4x_{10}) + 2h(0) = 0$$

$$2y_9 + (4h^2 - 2)y_{10} = 4h^2 x_{10}$$

The Tridiagonal System:

These equations form a linear system $AU = F$ of size $(m+1) \times (m+1)$. For $m = 10$, it looks like this:

$$\begin{bmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ 1 & 4h^2-2 & 1 & \dots & 0 & 0 \\ 0 & 1 & 4h^2-2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 4h^2-2 & 1 \\ 0 & 0 & 0 & \dots & 2 & 4h^2-2 \end{bmatrix} \begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_9 \\ y_{10} \end{bmatrix} = \begin{bmatrix} 0 \\ 4h^2x_1 \\ 4h^2x_2 \\ \vdots \\ 4h^2x_9 \\ 4h^2x_{10} \end{bmatrix}$$

- *Note:* The first row represents the Dirichlet BC, and the last row represents the Neumann BC (notice the 2 instead of a 1 in the lower diagonal).

Finite Difference Method: Stability & Convergence

- Numerical solution: $\mathbf{AU} = \mathbf{F}$
- Exact solution: $\mathbf{Au} = \mathbf{F} + \boldsymbol{\tau}$ ($\boldsymbol{\tau}$ is the local truncation error)

The Global Error:

- Subtracting yields Global Error $\mathbf{E} = \mathbf{u} - \mathbf{U}$:

$$\mathbf{AE} = \boldsymbol{\tau} \implies \mathbf{E} = \mathbf{A}^{-1}\boldsymbol{\tau}$$

- Taking matrix norms yields the fundamental convergence bound:

$$\|\mathbf{E}\| \leq \|\mathbf{A}^{-1}\|\|\boldsymbol{\tau}\|$$

The Condition for Stability:

- For central differences, $\|\boldsymbol{\tau}\| = O(h^2)$, but entries of $\mathbf{A}$ scale as $O(1/h^2)$.
- **Definition:** Stability requires $\|\mathbf{A}^{-1}\|$ to remain bounded by a constant C independent of h as $h \to 0$.

Stability Analysis: 1D Poisson Equation

Model Problem (1D Poisson):

$$-u''(x) = f(x) \quad \text{on } [0,1], \quad u(0) = 0,\ u(1) = 0$$

Central Difference Matrix A:

$$\mathbf{A} = \frac{1}{h^2} \begin{pmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \dots & 0 \\ 0 & -1 & 2 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 2 \end{pmatrix}_{(N-1)\times(N-1)}$$

$$\|\mathbf{E}\| \leq \|\mathbf{A}^{-1}\| \|\boldsymbol{\tau}\|$$

Spectral Norm: Definition & Error Bound

1. Why Spectral Norm ($\|\cdot\|_2$)?

- Measures maximum vector amplification in Euclidean space.
- Yields the fundamental global error bound:

$$\|\mathbf{E}\|_2 \leq \|\mathbf{A}^{-1}\|_2 \|\boldsymbol{\tau}\|_2$$

2. General Matrix Singular Value Definitions:

- Spectral norm of matrix $\mathbf{A}$:

$$\|\mathbf{A}\|_2 = \sigma_{\max}(\mathbf{A}) = \sqrt{\lambda_{\max}(\mathbf{A}^T\mathbf{A})}$$

- Spectral norm of inverse matrix $\mathbf{A}^{-1}$:

$$\|\mathbf{A}^{-1}\|_2 = \frac{1}{\sigma_{\min}(\mathbf{A})}$$

Spectral Norm of Real Symmetric Matrices

1. **Singular Values $\rightarrow$ Eigenvalues (when $\mathbf{A} = \mathbf{A}^T$):**
   - Since $\mathbf{A}^T\mathbf{A} = \mathbf{A}^2$, singular values simplify to eigenvalue magnitudes:

$$\sigma_i(\mathbf{A}) = \sqrt{\lambda_i(\mathbf{A}^2)} = |\lambda_i(\mathbf{A})|$$

2. **Eigenvalues of the Inverse Matrix $\mathbf{A}^{-1}$:**
   - If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, then multiplying by $\mathbf{A}^{-1}$ gives:

$$\mathbf{A}^{-1}\mathbf{v} = \frac{1}{\lambda}\mathbf{v}$$

3. **Final Inverse Norm Formula:**

$$\|\mathbf{A}^{-1}\|_2 = \max_i \left|\frac{1}{\lambda_i(\mathbf{A})}\right| = \frac{1}{\min_i |\lambda_i(\mathbf{A})|} = \frac{1}{\lambda_{\min}(\mathbf{A})}$$

Eigenvalues of the 1D Poisson Operator

- **Analytical Eigenvalues of A:**

$$\lambda_k = \frac{2}{h^2}\left(1 - \cos(k\pi h)\right) \quad \text{for } k = 1, 2, \ldots, N-1$$

- **Smallest Eigenvalue ($k = 1$):**

$$\lambda_1 = \frac{2}{h^2}\left(1 - \cos(\pi h)\right)$$

- **Taylor Expansion ($\cos(\pi h) \approx 1 - \frac{\pi^2 h^2}{2}$):**

$$\lambda_1 \approx \frac{2}{h^2}\left(1 - \left(1 - \frac{\pi^2 h^2}{2}\right)\right) = \pi^2$$

- **Asymptotic Inverse Norm Ceiling ($h \to 0$):**

$$\|\mathbf{A}^{-1}\|_2 = \frac{1}{\lambda_1} \approx \frac{1}{\pi^2}$$

Final Convergence & Physical Meaning

The Global Error Bound:

- Truncation Error: $\|\boldsymbol{\tau}\|_2 = O(h^2)$ (from 2nd-order central differences)
- Stability Ceiling: $\|\mathbf{A}^{-1}\|_2 \leq \frac{1}{\pi^2}$ (bounded independent of h)

$$\|\mathbf{E}\|_2 \leq \|\mathbf{A}^{-1}\|_2 \|\boldsymbol{\tau}\|_2 \leq \frac{1}{\pi^2} \cdot O(h^2)$$

Takeaways:

- **Convergence:** As $h \to 0$, $\|\mathbf{E}\|_2 \to 0$. The discrete numerical solution strictly approaches the true continuous solution $u(x)$.
- **Stability:** Even though individual matrix entries scale up as $O(1/h^2)$, $\|\mathbf{A}^{-1}\|_2$ is bounded above by the constant $1/\pi^2$, independent of grid spacing h.
- Grid refinement will not blow up truncation or roundoff errors.

Terminology and Classification of Linear 2nd-Order PDEs

General 2nd-Order Linear PDE

$$\underbrace{A(x,y)\frac{\partial^2 u}{\partial x^2} + B(x,y)\frac{\partial^2 u}{\partial x \partial y} + C(x,y)\frac{\partial^2 u}{\partial y^2}}_{\textbf{Principal Part}} + \underbrace{D(x,y)\frac{\partial u}{\partial x} + E(x,y)\frac{\partial u}{\partial y} + F(x,y)u}_{\textbf{Lower-Order Part}} = g(x,y)$$

- **Principal Part (2nd-Order Derivatives)**
  - Governed by coefficients $A(x,y)$, $B(x,y)$, and $C(x,y)$.
  - Captures the highest spatial/temporal rates of change.
- **Lower-Order Part (1st-Order Derivatives & Variable u)**
  - Governed by coefficients $D(x,y)$, $E(x,y)$, $F(x,y)$, and forcing $g(x,y)$.
  - Represents secondary physical processes (e.g., advection drift, damping, source terms).

The Discriminant and Coordinate Invariance

For the general linear 2nd-order PDE, the **discriminant** is defined as:

$$D = B^2 - 4AC$$

Key Property:

- The sign of the discriminant $B^2 - 4AC$ is **always invariant** under any non-singular linear transformation of independent variables:

$$(x, y) \longrightarrow (\xi, \eta)$$

- Physical behavior is intrinsic to the equation and does not depend on the choice of coordinate system.

A 2nd-order linear PDE can be uniquely classified according to the sign of its discriminant.

Classification of 2nd-Order Linear PDEs

- **1. Elliptic** $(B^2 - 4AC < 0)$
  - Describes steady-state / equilibrium processes.
  - Ex: Poisson/Laplace

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

- **2. Parabolic** $(B^2 - 4AC = 0)$
  - Describes dissipation and diffusion processes over time.
  - Ex: 1D Heat Equation

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

- **3. Hyperbolic** $(B^2 - 4AC > 0)$
  - Describes wave propagation and dynamic transport.
  - Ex: 1D Wave Equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Why Classification is Important

- **Governs Information Propagation**
  - Dictates characteristic curves and domains of dependence.
  - Determines whether signals propagate instantaneously (Elliptic/Parabolic) or at finite speeds (Hyperbolic).
- **Guides Numerical Strategy**
  - Determines necessary boundary/initial conditions for a well-posed problem.
  - Selects stable numerical schemes (e.g., implicit solvers for Elliptic/Parabolic vs. constrained explicit schemes for Hyperbolic).
- **Lower-Order Invariance**
  - Adding drag, damping, or source terms introduces physical variation in values, but **never** alters the underlying PDE classification.